

Introducing RebelTrack and RebelCast

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Predictive turn in conflict studies

- ▶ Gurr and Lichbach, 1986
- ▶ O'brien, 2002
- ▶ Schrod, 2006
- ▶ Goldstone et al., 2010
- ▶ Ward, 2010
- ▶ Weidmann and Ward, 2010
- ▶ Schneider, Gleditsch and Carey, 2011
- ▶ De Mesquita, 2011
- ▶ Ward et al., 2013
- ▶ Gleditsch and Ward, 2013
- ▶ Hegre et al., 2013
- ▶ Bell et al., 2013
- ▶ Brandt, Freeman and Schrod, 2014
- ▶ Muchlinski et al., 2016

Advances and limitations

- ▶ More open-source data
- ▶ Better statistical models
- ▶ Feature development compartmentalized
- ▶ Little bench-marking and comparison

- ▶ Observation: Open-source data availability more established in our field than open-source software development
- ▶ R package for measuring rebel organizations' behavior, relationships, and characteristics.
 - ▶ Version 0.8 mainly based on UCDP GED (Sundberg and Melander, 2013) and PRIO-GRID dataset (Tollefsen, Strand and Buhaug, 2012)
 - ▶ Focus on actors that actually make decisions about conflict
 - ▶ Open-source development effort on Github
 - ▶ Developers gain authorship in yearly update
 - Contributor
 - Provider
 - Coordinator
 - ▶ Currently software can generate 146 indicators
 - ▶ Data generation and data provision comply with the Data Access and Research Transparency (DA-RT) initiative
 - Open code, data (download date, version), codebook etc.

Mean fighting position

side_b:

Sendero Luminoso ▼

Data from UCDP GED



Figure: Shiny interface to explore RebelTrack data. Users can choose a specific rebel organization and the respective measurement. Examples provided for Sendero Luminoso in Peru. Top panel: Mean fighting position of rebel organizations, where darker dots represent more recent monthly mean fighting positions. Middle panel: Number of PRIO-grids that are affected by rebel-government fighting in a particular month. Bottom panel: Mean distance of fighting events to the capital.

PRIO Grid count

side_b:

Sendero Luminoso ▼

Data from UCDP GED

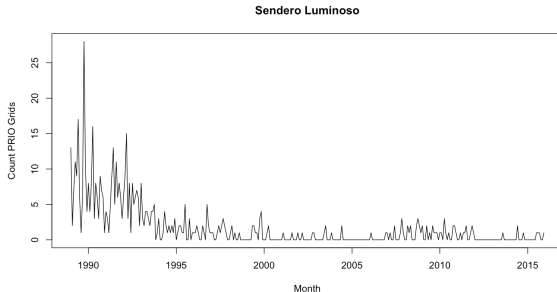


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Distance to capital

side_b:

Sendero Luminoso ▼

Data from UCDP GED

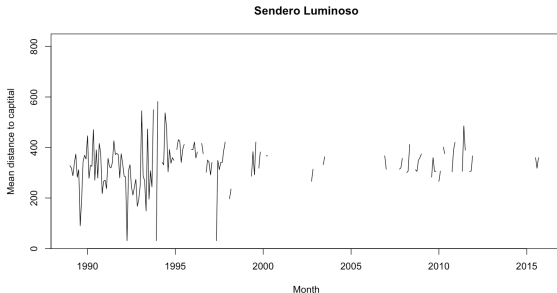


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- ▶ RebelTrack allow users to generate monthly, quarterly, or yearly aggregations
- ▶ Users can also choose whether variables should be lagged and by how many time units.
- ▶ Option to perform missing data imputation is provided
- ▶ Release plan:
 - ▶ RebelTrack version 1.0, release early Winter 2017: Rebel behavior
 - ▶ RebelTrack version 1.1, release late Winter 2017: Rebel dependencies and configurations
 - ▶ RebelTrack version 1.2, release early Spring 2018: Rebel characteristics
 - ▶ RebelTrack version 1.3, release late Spring 2018: Inclusion of event data (similar to MELTT and geomerge)

- ▶ Observation: Many individual forecasting approaches, little bench-marking and comparison.
- ▶ Open source framework for predicting conflict dynamics (escalation, **duration**)
 - ▶ Early warning.
 - ▶ Machine Learning approaches help to improve sensitivity-specificity tradeoff (true-positives/true-negatives) (Muchlinski et al. 2016)
 - ▶ Ensemble Bayesian Model Averaging models help us to integrate the predictions from several models (Montgomery, Hollenbach, and Ward, 2012)
 - ▶ Allows users to add their own models to the ensemble

RebelCast Duration

When does the fighting stop?

Machine learning approach

- ▶ Outcome: End of UCDP Rebel Spells (Kreutz, 2010)
- ▶ Unit of analysis: Quarter year - rebel organization
- ▶ 278 rebel organizations
- ▶ Observed in 9377 quarters
- ▶ Group-wise cross-validation
- ▶ Eight Machine learning algorithms
- ▶ Averaged with an Ensemble Model
- ▶ Input
 - ▶ RebelCast variables lagged by one quarter
 - Mean Capital distance, Border distance, SideB casualties, SideA casualties, PRIO grids affected, Transnational fighting count, Transnational fighting ratio, Resource variables (Diamonds, Drugs, Gold, Oil), Terrain variables (urban, mountainous, forest, and other terrain types)

- ▶ **Logit:** Standard logistic regression model implemented in the R stats package.
- ▶ **Random Forest:** Classification algorithm based on tree classification. Implemented in the R package randomForest.
- ▶ **Weighted Subspace Random Forest for Classification:** Implemented in the R package wsrf. This algorithm optimizes the selection of feature subspaces. Instead of randomly selecting variables to build random forest models, this model excludes non relevant features in the sampling process.
- ▶ **AdaBoost:** This algorithm was developed by Freund and Shapire, 1996. It leverages boosting to increase accuracy and is implemented in the R package fastAdaboost.
- ▶ **gbm:** Generalized Boosted Regression Modeling, which extends the Freund and Shapire, 1996 algorithm. This is implemented in the R package gbm.
- ▶ **svmRadial:** Support Vector Machine implemented in the R kernlab package.
- ▶ **knn:** K-nearest neighbour: Units are classified according to their k-nearest neighbors is a particular feature space using the class

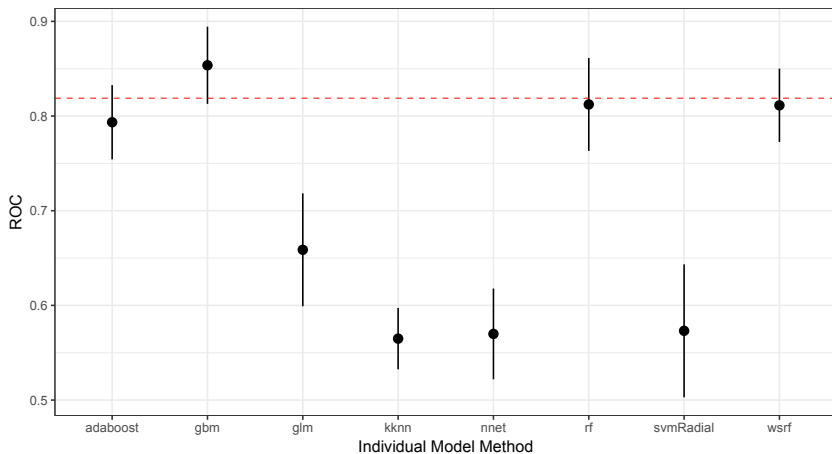


Figure: Area under the Curve for all eight statistical learning methods that are included in the ensemble model. Red dotted line represents the average ensemble AUC. Results for 10-fold groupwise cross-validation.

Ensemble

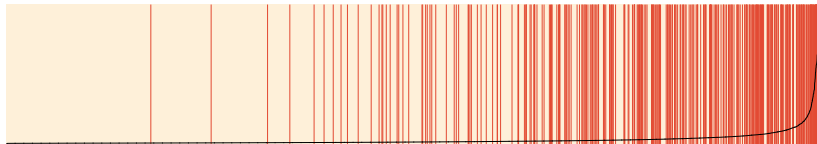


Figure: Separationplot from ensemble model ranking observed termination quarters (red) and observed continuation quarters (white) according to their predicted probability from left (low values) to righting (high values)

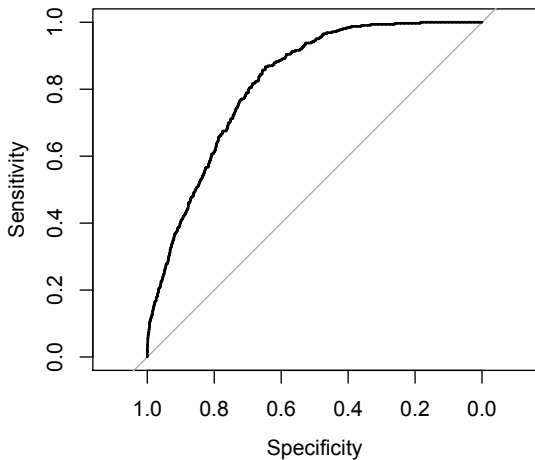


Figure: ROC plot showing the tradeoff between sensitivity and specificity of the ensemble model.

Table: Confusion matrices for ensemble model with different classification thresholds. Left confusion matrix with classification threshold = 0.5. Right confusion matrix with the mean of all predicted probabilities as threshold.

	Pred. = 0	Pred. = 1
Obs. = 0	9032	22
Obs. = 1	303	20

(A): Classification threshold=0.5

	Pred. = 0	Pred. = 1
Obs. = 0	7014	2040
Obs. = 1	105	218

(B): Classification threshold=Mean

Conclusion and looking ahead

- ▶ Success dependent on incentives to contribute
- ▶ Hope to contribute to cumulative indicator development and forecasting
- ▶ EBMA allows to integrate and weigh predictive strengths of different models